Color Equivariant Convolutional Networks [1]

Color is a crucial visual cue for object recognition, but CNNs fail to generalize if there is data imbalance between color variations. Color invariance addresses this issue by removing all color information. We introduce Color Equivariant Convolutions (CEConvs): a novel deep learning building block that shares shape features across the color spectrum while retaining important color information. [2]

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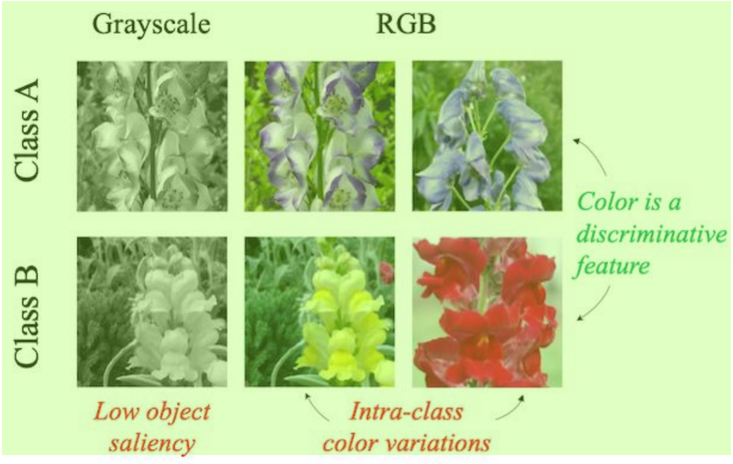


[6]

Figure 1. Color is often a discriminative feature in object recognition. However, intra-class color variations can confuse a classification model. On the other hand, removing color makes flowers less distinct from their background and thus harder to classify. [7]

01 Preliminaries [8]

A CNN is *equivariant* to a transformation *T* if transforming the input *x* by *T* results in an equally transformed feature map: [11]

$$f(T(x)) = T'(f(x))$$ [12]

Equivariance allows *parameter sharing*, resulting in better *data efficiency* and *OOD generalization*. [13]

Transformation information is stored in extra FM dimension – *invariance* achieved by avg pooling. [14]

02 Contributions [9]

1. We show that CNNs *benefit from color information* and at the same time are *not robust to color shifts*. [15]
2. We introduce Color Equivariant Convolutions (CEConvs), which allows *feature sharing between colors*.
3. We demonstrate that CEConvs improve robustness to train-test color shifts in the input.

03 Color Equivariant Convolutions [10]

- CEConvs implement equivariance to *hue shifts*. [16]
- Hue shifts are modeled as 3D rotations around [1,1,1] diagonal, parameterized as a 3x3 rotation matrix *H*.
- CEConv input layer is defined as:

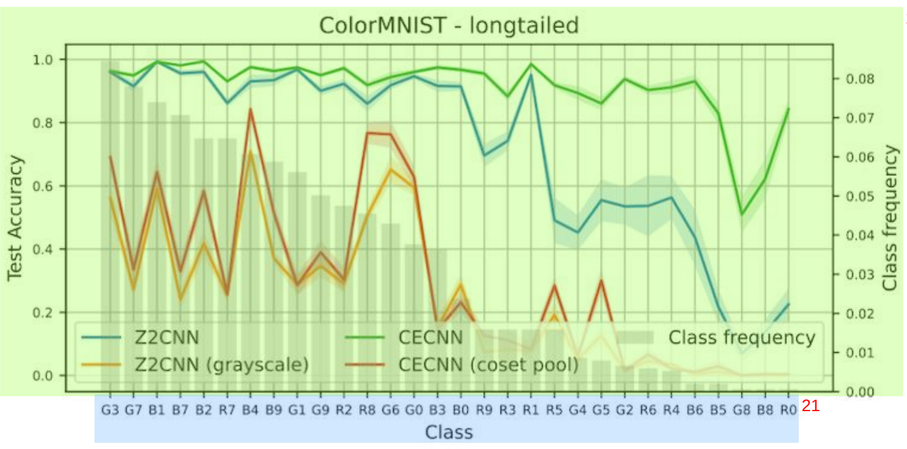
$$[f \star \psi^i](x,k) = \sum_{y \in Z^2} \sum_{c=1}^{C^l} f_c(y) \cdot H_n(k) \psi_c^i(y - x)$$

- Hidden layers: cyclic permutations in color dimension.
- Hybrid CECNNs use CEConvs in part of network.

04 When is color equivariance useful? [17]

Synthetic toy settings: [18]

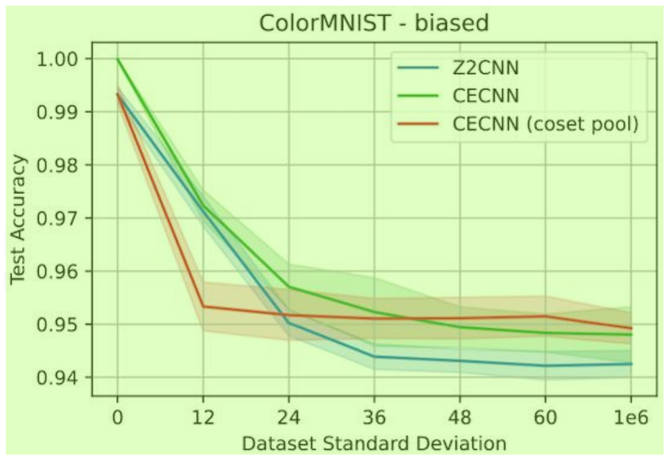
1. Color *imbalance*: simulated by *long-tailed ColorMNIST*, a 30-class classification problem where digits need to be classified by number (0-9) and color (red, green, blue). [19]
2. Color *variations*: simulated by *biased ColorMNIST*, a 10-class classification problem where each class *c* has its own characteristic hue θ_c defined in degrees, distributed uniformly on the hue circle. The exact color of each digit *x* is sampled according to $\theta_x \sim N(\theta_c, \sigma)$.



[20] [21]

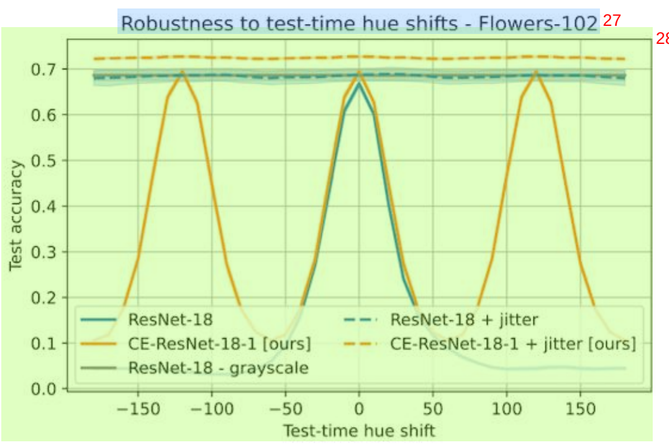
< Left *Long-tailed ColorMNIST - CECNN (91.35 ± 0.40%) performs significantly better than Z2CNN (71.59 ± 0.61%). Performance mostly increases for classes with few samples, indicating that CEConvs are indeed more data efficient. Z2CNN (grayscale) and CECNN (coset pool) with an average accuracy of 24.19 ± 0.53% and 29.43 ± 0.46%, respectively, are unable to discriminate between colors.* [22]

Right > *Biased ColorMNIST - Test accuracies for different standard deviations (σ). CECNN outperforms Z2CNN across all σ. CECNN (coset pool) outperforms CECNN for σ ≥ 48 - above this value color is no longer informative and acts as noise. Z2CNN (grayscale) is omitted as it performs significantly worse, ranging between 89.89% (σ = 0) and 79.94 (σ = 10⁶).* [23]



[24]

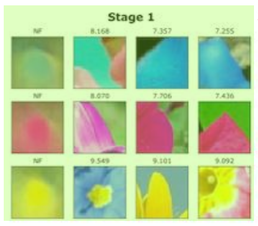
05 Robustness to color shifts [25]



[27] [28]

< Left *Test accuracies under gradual test-time hue shift. Hue variations degrade the performance of CNNs (ResNet-18). Grayscale images and color jitter are invariant and thus robust, but also fail to capture useful color features. Our color equivariant network (CE-ResNet-18-1) enables feature sharing across colors and generalizes to discrete hue shifts. Equivariance is complementary to color jitter as the combination performs best.* [30]

Right > *Neuron Feature visualizations of filters at different depths in the network. Rows represent the color dimension and encode the same shape in different colors.* [31]



[32]

[33]

06 Conclusions [26]

- Color Equivariant convolutions *enable feature sharing across colors* and, unlike invariance, can still use color information. [34]
- CECNNs outperform regular CNNs in a color imbalanced or biased setting.
- CEConvs are computationally *more expensive* than regular convolutions. *Hybrid CECNNs* provide early equivariance with only limited compute increase.
- CEConvs are only *approximately equivariant* due to clipping errors.
- Future work: combine *color and geometric transformations* in equivariant convolutions.

Related Literature [29]

- *Group equivariant convolutional networks - Taco S. Cohen and Max Welling, in the 33rd International Conference on Machine Learning, 2016*
- *Color encoding in biologically-inspired convolutional neural networks - Ivet Rafegas and Maria Vanrell, in Vision Research, 2018.*

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